



Multimodal Data Integration for 30-Day Mortality Prediction in Hemorrhagic Stroke: The PREDICH Project

Presented by: Amine Sadikine, Postdoctoral Researcher, University Hospital of Brest | Inserm, UMR 1078, GGB

Principal Investigator: Prof. Serge TIMSIT, University Hospital of Brest | Inserm, UMR 1078, GGB

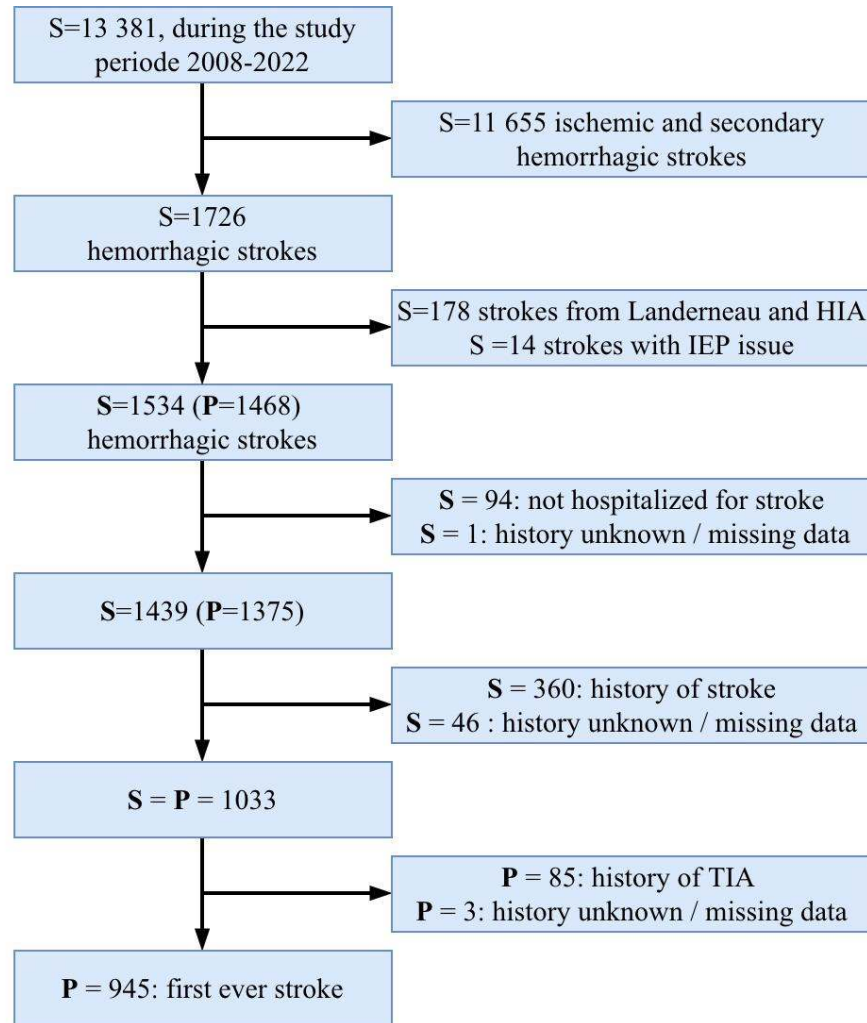
Institution: University Hospital of Brest (CHU), University of Western Brittany (UBO)

Date: January 27, 2026

Event: STRATIF-AI annual meeting

Study Population Selection

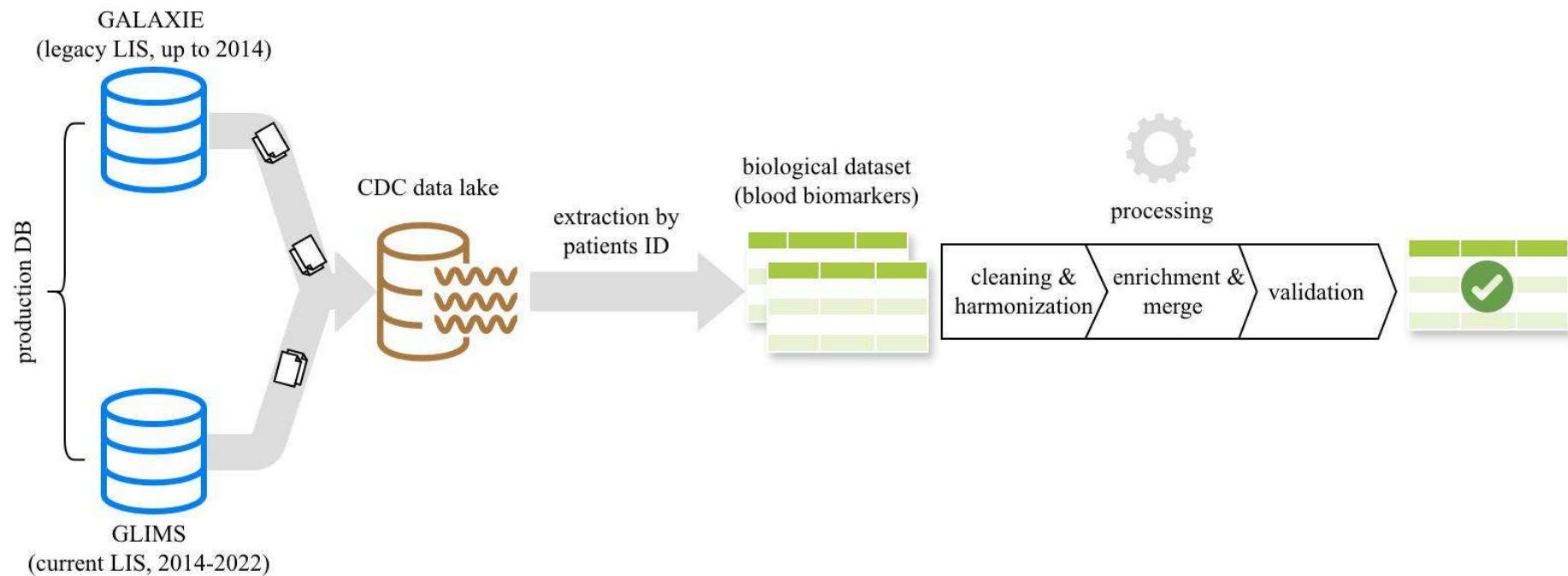
S: number of stroke
P: number of patients



Blood Biomarkers

Clinical Category (12)	Biomarkers (64)
Ionogram & Renal Function	Sodium, Potassium, Chloride, Bicarbonate, Urea, Creatinine, Urea/Creatinine Ratio, Calcium, Corrected Calcium, Phosphorus, Magnesium, MDRD, CKD-EPI
Nutritional Markers	Albumin, Plasma Proteins
Lipid Profile	Total Cholesterol, Triglycerides, HDL Cholesterol, Calculated LDL Cholesterol
Diabetes Markers	Glucose, Glycated Hemoglobin / HbA1c
Cardiac Markers	BNP, NT-proBNP, Troponin I
Muscle Markers	CPK
Inflammatory Markers	CRP (C-Reactive Protein)
Liver Panel	Total Bilirubin, Alkaline Phosphatase, Gamma GT, ALT, AST
Iron & Anemia Panel	Iron, Ferritin
Thyroid Function	Ultra-sensitive TSH
CBC (Complete Blood Count)	Red Blood Cells, Hemoglobin, Hematocrit, MCV, MCH, MCHC, RDW, Platelets, MPV, Leukocytes, Neutrophils, Eosinophils, Basophils, Lymphocytes, Monocytes
Coagulation & Hemostasis	ESR (VS), Quick Time (Control/Patient), Prothrombin Time, aPTT (Patient/Ratio), Fibrinogen, D-Dimer

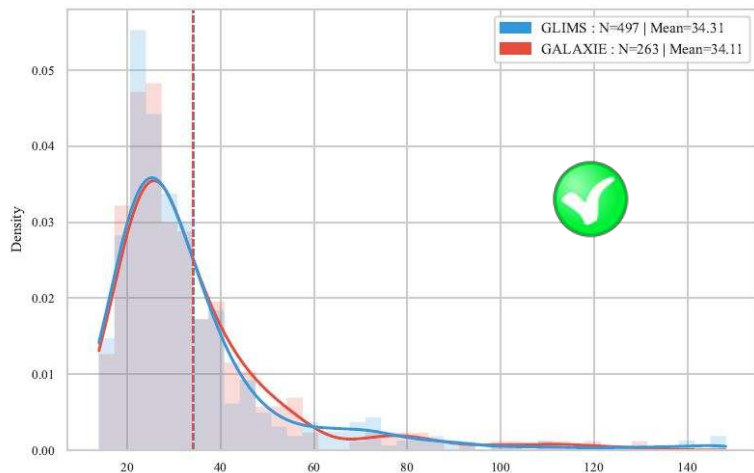
Biological Data Retrieval and Processing Pipeline



*LIS: Laboratory Information System

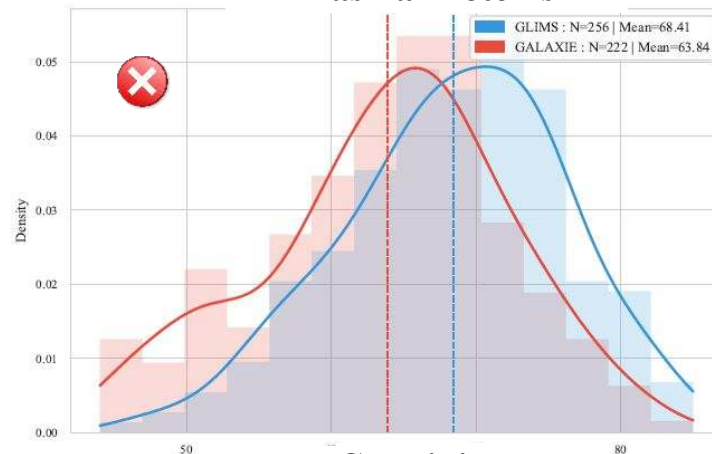
Inter-Source Distribution Comparison

AST

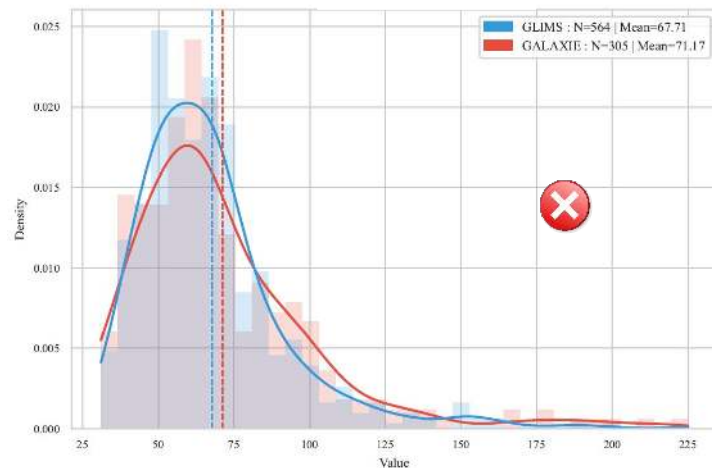


→ 30% of biomarkers show significant distribution shifts between sources.

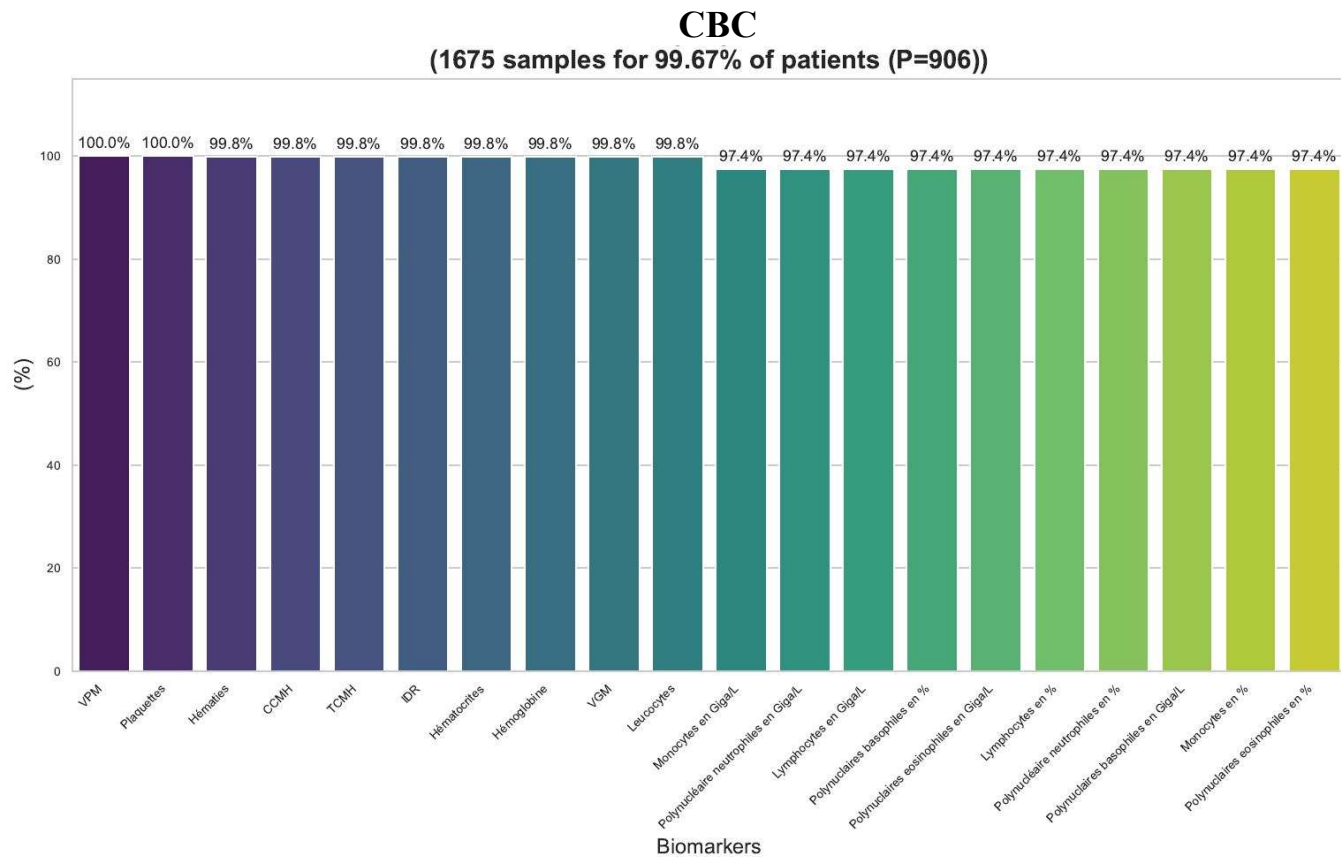
Plasma Proteins



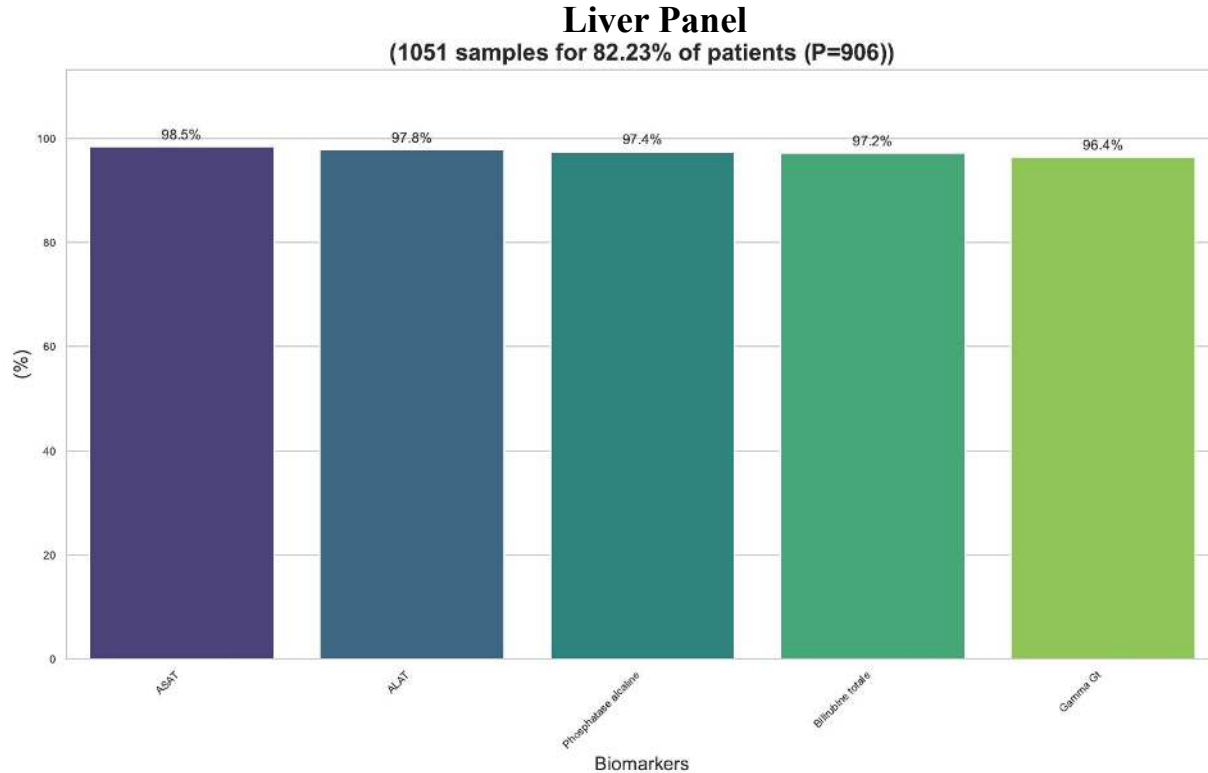
Creatinine



Biomarker Availability within the First 48 Hours (1/3)



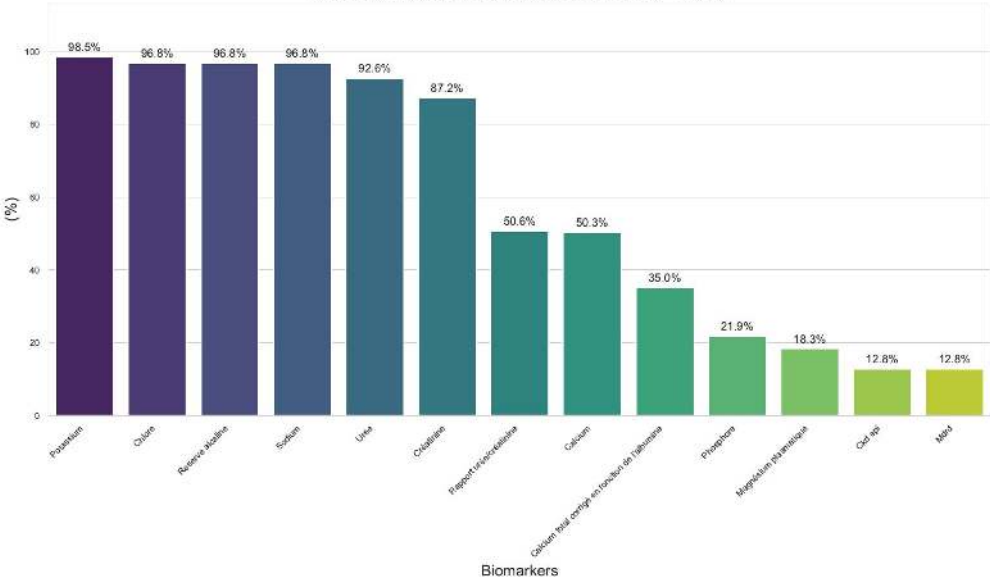
Biomarker Availability within the First 48 Hours (2/3)



Biomarker Availability within the First 48 Hours (3/3)

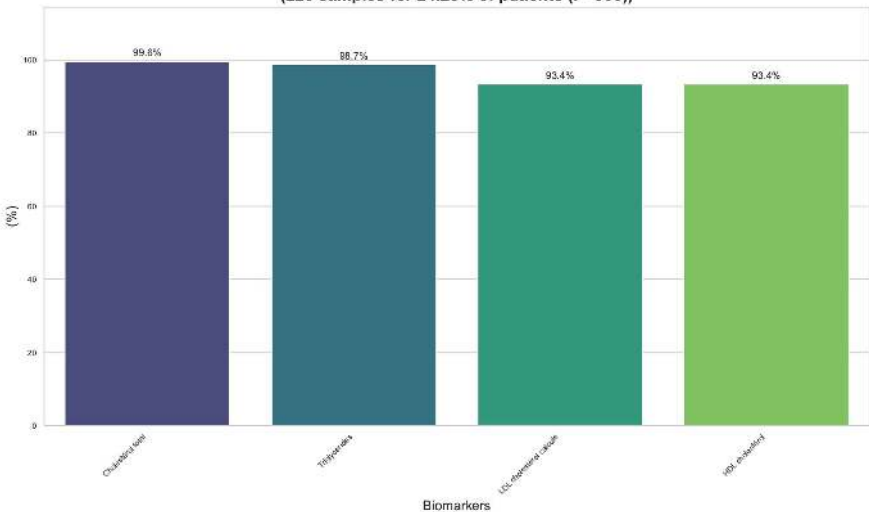
Ionogram

(2001 samples for 99.67% of patients (P=906))



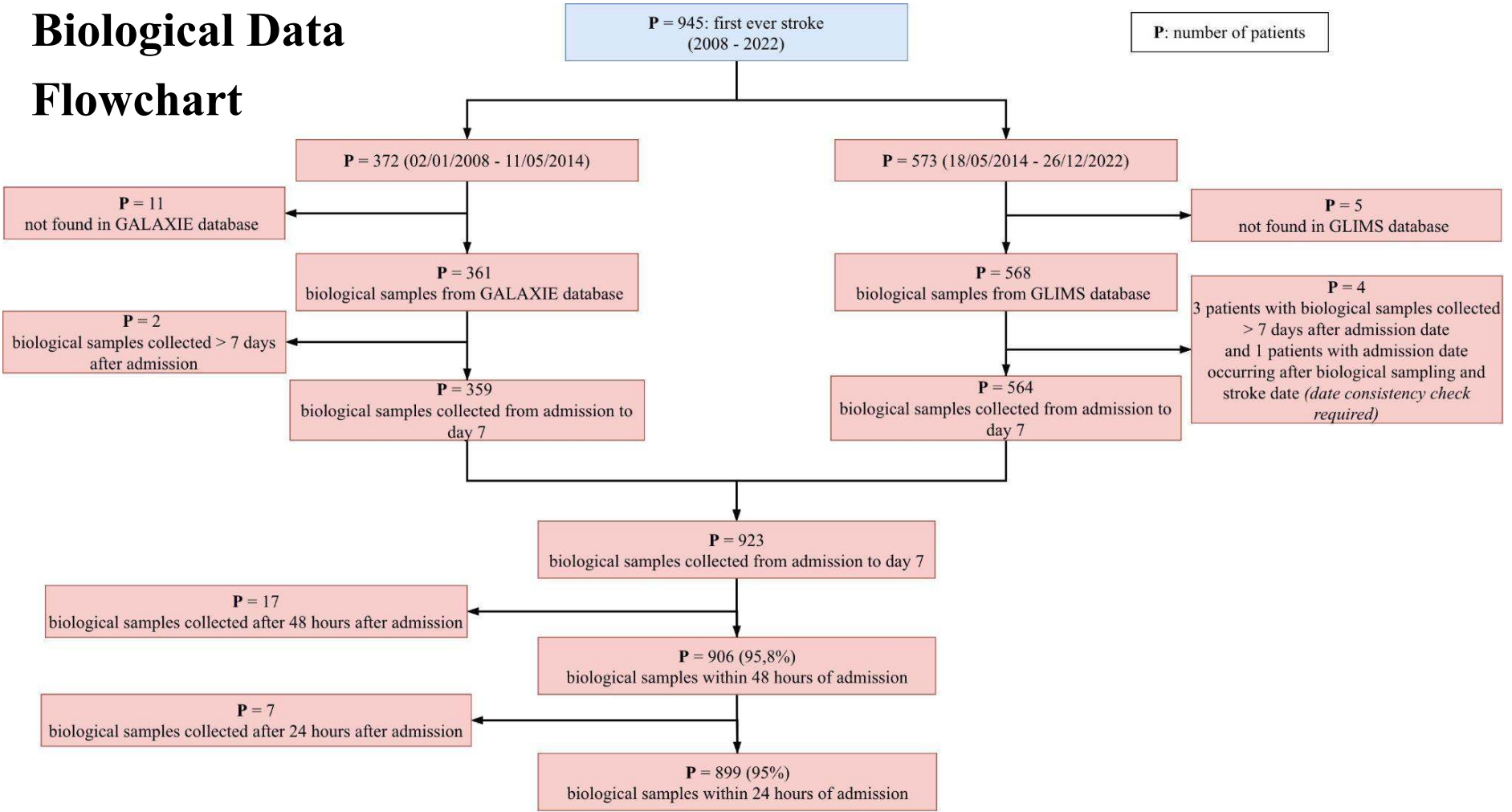
Lipid Profile

(226 samples for 24.28% of patients (P=906))

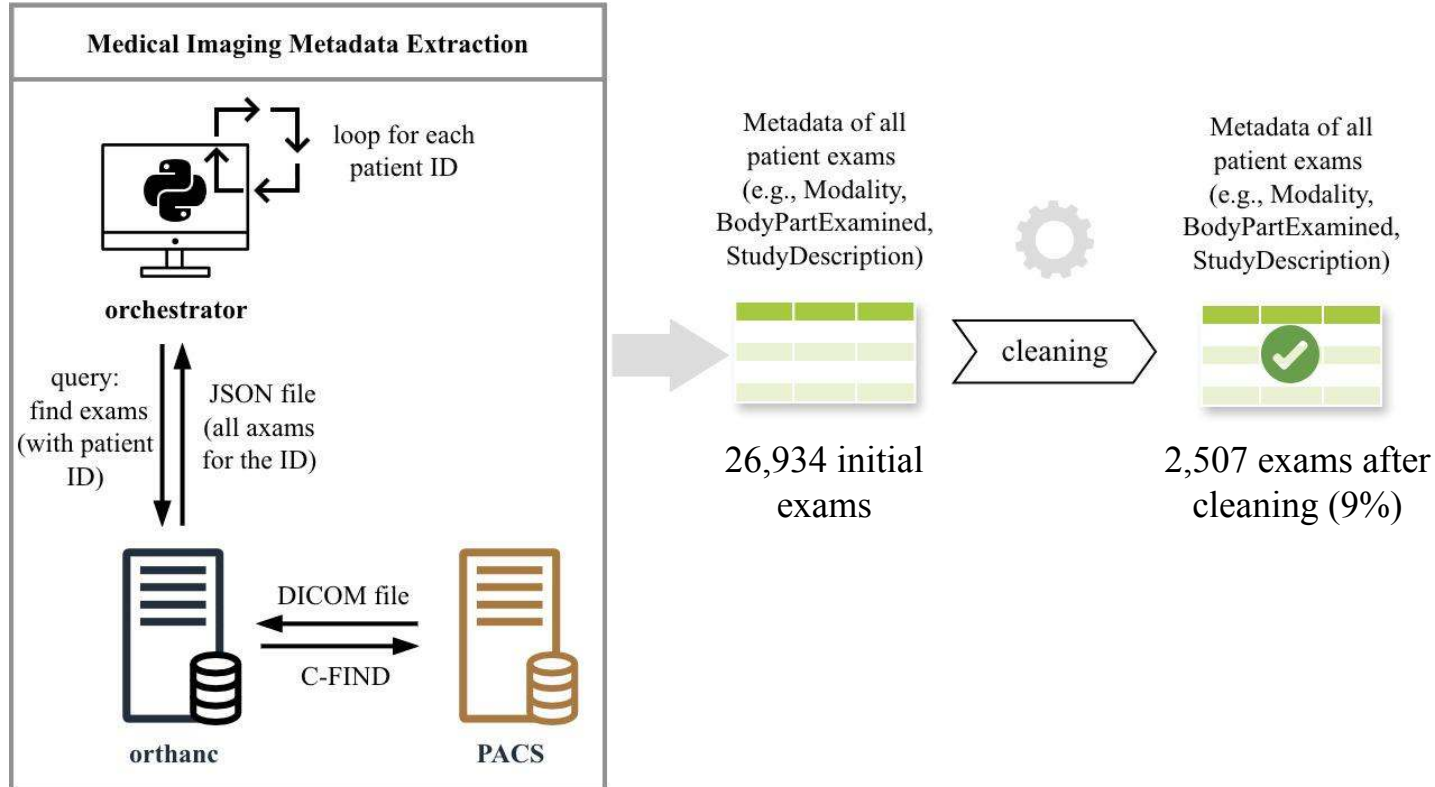


Biological Data

Flowchart

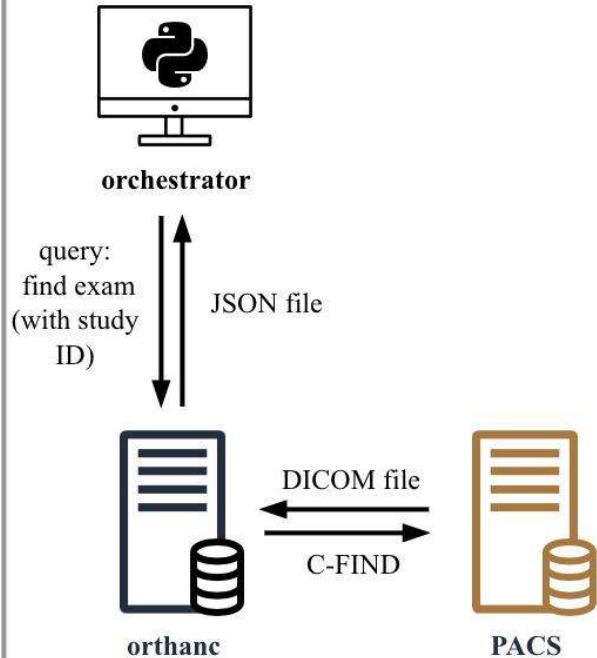


CT Scan Metadata Retrieval & Processing Pipeline (1/2)

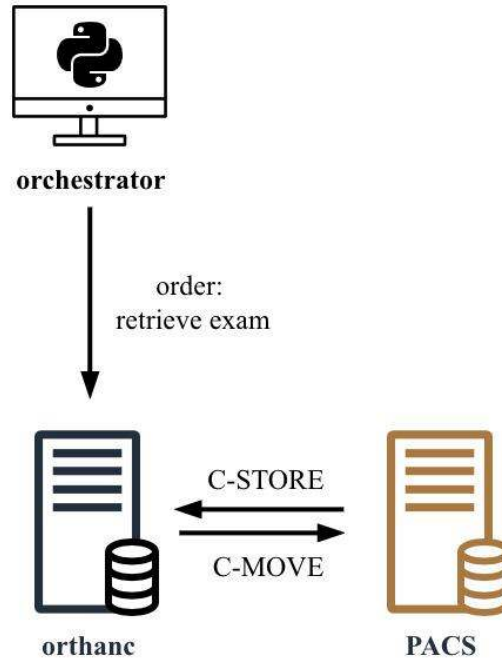


Medical Exam Retrieval Process (2/2)

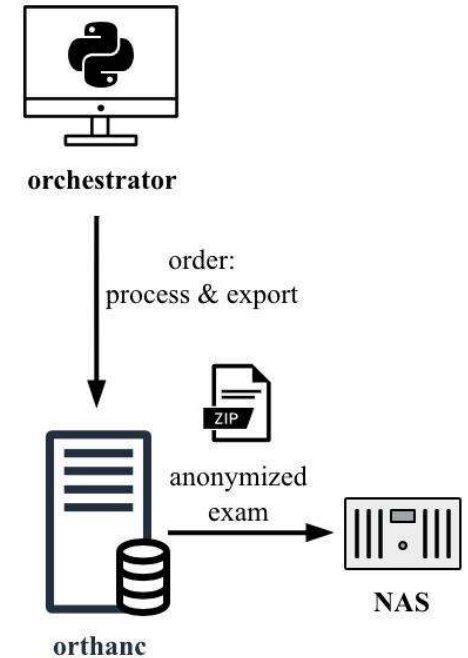
PHASE 1: Query (C-FIND)



PHASE 2: Retrieve (C-MOVE & C-STORE)



PHASE 3: Anonymization & Export

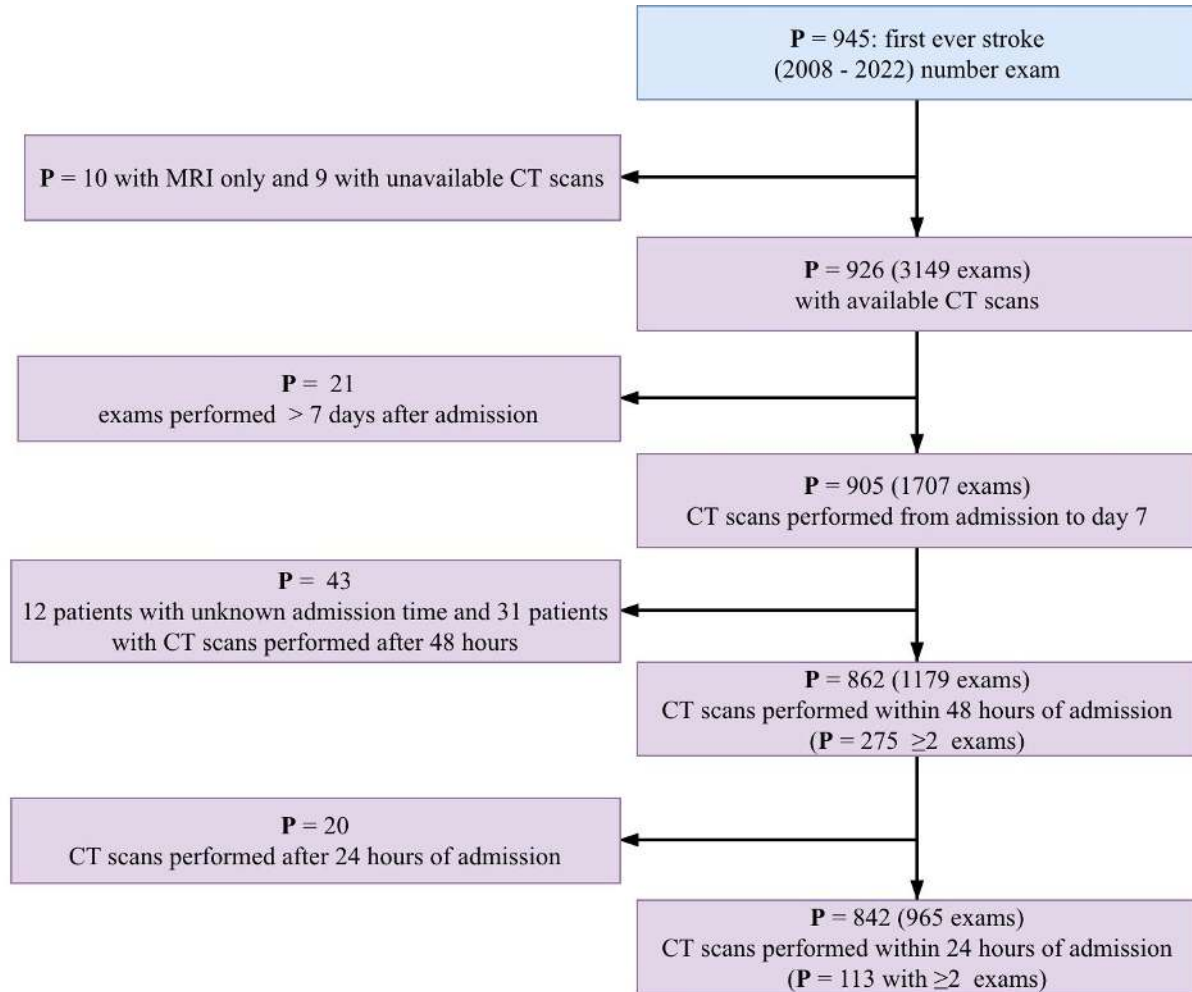


Imaging Data Status: Retrieval Progress & Next Steps

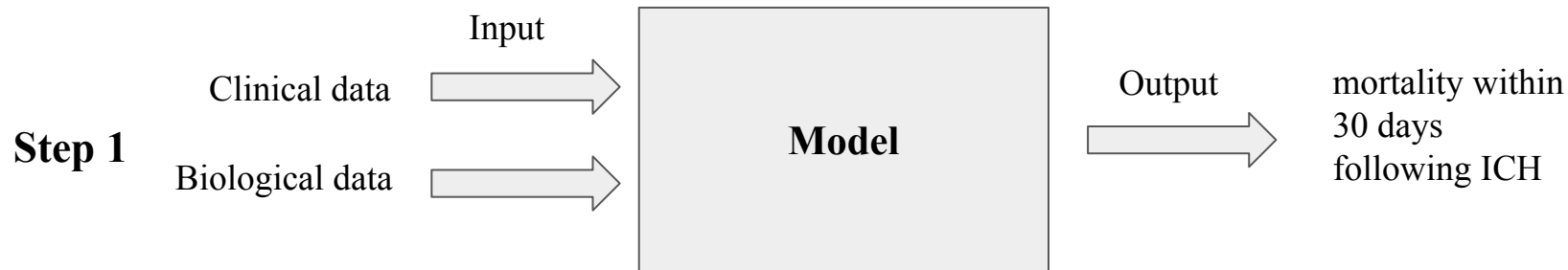
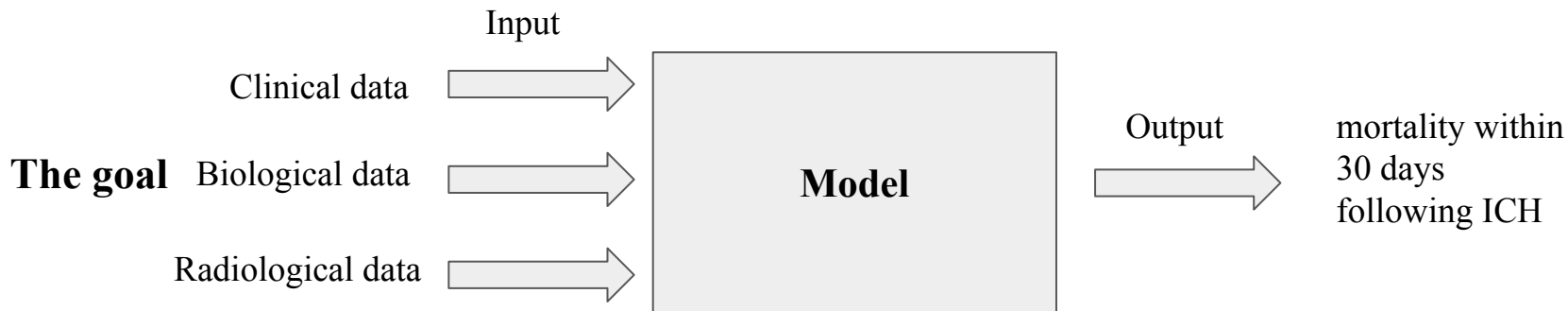
- **CT Scans Status:** 925 / 2,507 exams retrieved (1,323 patients) and 1,582 remaining downloads in progress.
- **Imaging Pipeline:** Post-retrieval steps include data cleaning, NIfTI conversion, and final validation.
- **Upcoming MRIs:** 1,308 MR exams identified (599 patients) for future retrieval.

CT Imaging Flowchart

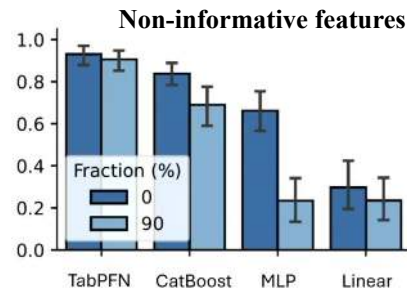
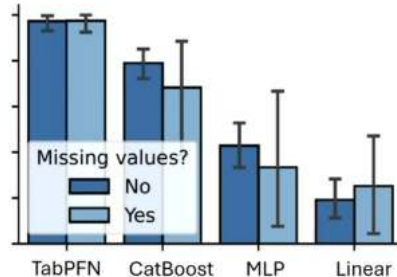
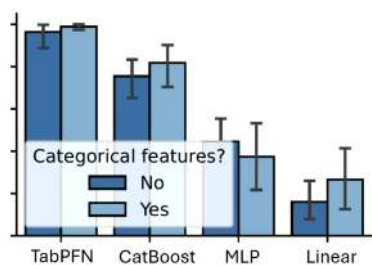
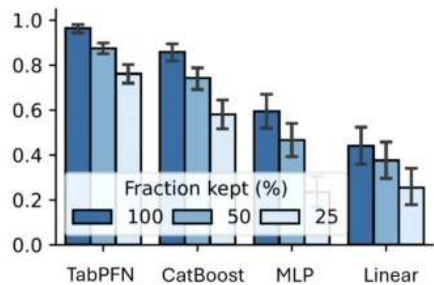
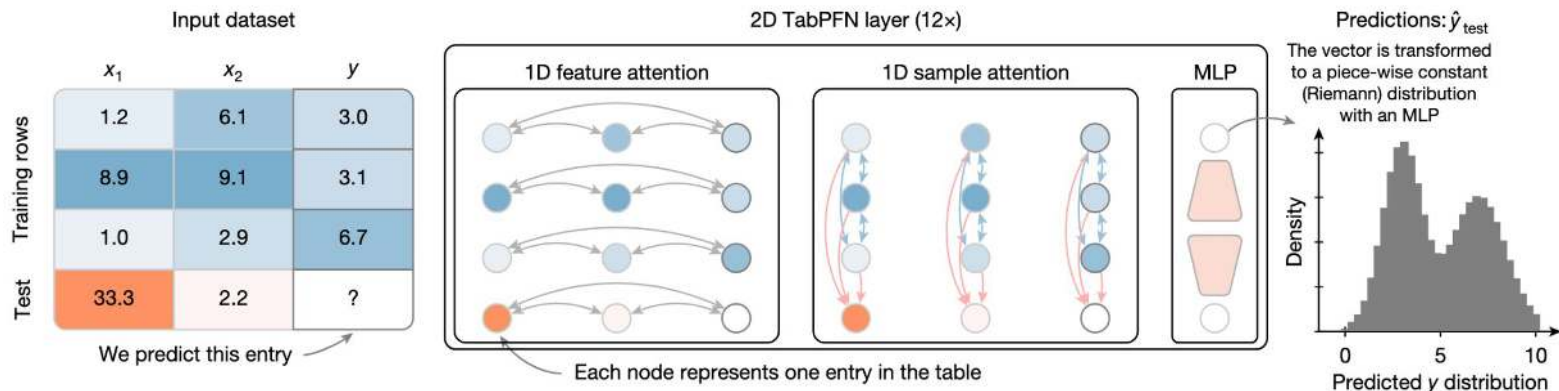
P: number of patients



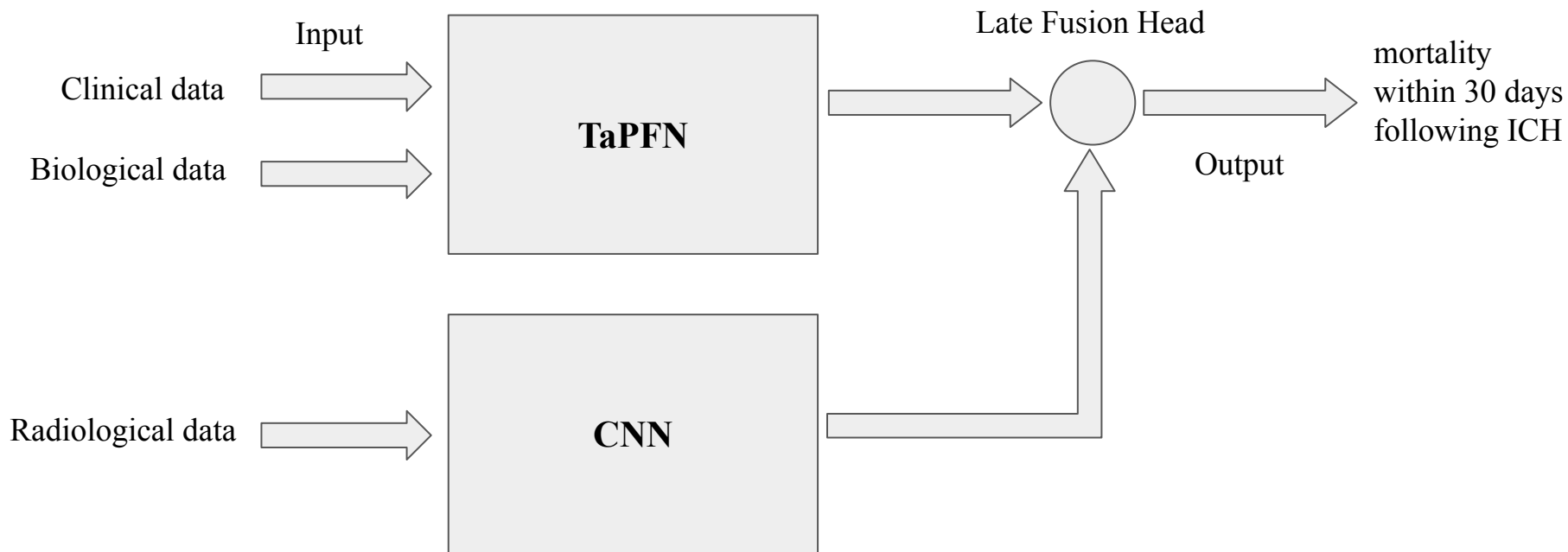
Multimodal Prediction Of 30-Day Mortality



Tabular Data Modeling: TabPFN



Integration of Imaging



Conclusion & Perspectives

Multi-Source Data Integration: Successful consolidation of 64 unique biomarkers from two distinct LIS sources (GALAXIE and GLIMS) over the period 2008–2022.

Data Completeness: Achieved a high completeness rate of 97% for CBC biomarkers within the critical 48-hour window.

Imaging Pipeline: Progressed in the automated retrieval of CT scans with 925 exams already secured and 1,582 additional downloads currently in progress.

Proposed Multimodal Architecture: Designed a predictive framework to integrate clinical, biological and radiological data.

Validation Roadmap: Final steps involve completing the internal model training on the primary dataset followed by external validation to verify 30-day mortality risk stratification.

Imaging Expansion: Upcoming milestones include the retrieval and processing of 1,308 identified MR exams for the future work.

Thanks for your attention!